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(1) Research Program/Project/Study Title: EFFECT OF SERPENTINA (*Andrographis paniculata*) PLANT POWDER IN THE IMMUNE RESPONSE TO NEWCASTLE DISEASE OF DARAG NATIVE CHICKEN (*Gallus gallus domesticus*)

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Program or Project has component (if no proceed to 2)? ☐ Yes ☒ No

Component Title:	
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Academic Rank:	

(2) Category (please check)

- ☐ Social Science/Education /Humanities & other related fields
☒ Natural/Biological Sciences
☐ Food Research/
☐ Developmental/ Computer/Engineering/ Architecture
☐ Extension
☐ Poster

(3) Research Center

[Check the concerned center]

- ☐ SSITRDC
 Social Science and Industrial Technology Research Development Center

☒ LRDC
 Livestock Research Development Center

☐ FFRDC
 Fisheries and Food Research Development Center

☐ CSRDC
 Crop Science Research Development Center

(4) Site of implementation

Municipality/City:	Dumarao
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Region: VI	
(5) Classification Research: Development: ☒ Basic ☐ Pilot Testing ☐ Applied ☐ Tech. Promotion/Commercialization	(6) Mode of Implementation ☒ Single Authorship/Proponent ☐ Collaborative Research (within campus) ☐ Collaborative Research (intercampus) ☐ Collaborative Research (external agency)
(7) Sector/Commodity Livestock	(8) Discipline/Field of Specialization Native Chicken
(9A) Abstract Newcastle disease is one of the most important diseases of chickens that can lead to economic losses of raisers. Vaccine is the only remedy to prevent this disease. To address this gap, the study was conducted to evaluate the effect of powdered serpentina plant on the immune response to Newcastle disease of Darag native chickens at day 2, 4, 8, and 12 weeks post treatment. The study also aims to evaluate the effects of powdered serpentina plant in the liver of native chicken based on their SGPT and SGOT level and in the productive performance of Darag native chickens. Factorial experiment was employed in this study where Factor A are the parts of serpentina plant powder and commercial drugs and Factor B are the dosages of 1 g/kg BW, 1.5 g/kg BW and 2 g/kg BW of the experimental birds. The study used a total of two hundred twenty-five (225) heads of Darag native chickens of both sexes' ages ranging from 2-4 months old and arranged in a 5 x 3 factorial experiment in a Completely Randomized Design (CRD). Data gathered were transformed and analyzed using ANOVA. Treatment means were compared using DMRT and were interpreted at 5% and 1% level of significance. The study found that administration of Serpentina plant powder at various levels had no effect on the Newcastle disease titer in Darag native chickens at 2, 4, 8, and 12 weeks post-treatment. Serpentina plant powder can decrease SGPT and SGOT level of chickens that shows of a healthy liver for the chickens. Data further revealed that chickens given with serpentina plant powder has a significant effect in the gain in weight and feed conversion ratio of chickens similar to commercial drugs.	
(9B) Keywords: <i>Feed consumption, Feed conversion ratio, Serum glutamate pyruvate transaminase, serum glutamic-oxaloacetic transaminase</i>	
(10) Introduction The role of native chicken in Philippine agriculture and the entire economy is well-recognized. It will, to a large extent, remain a significant contributor to the continuous supply of meat and eggs and extra income for many rural Filipino farmers. Its meat is highly preferred by many Filipino consumers because of its distinct taste, leanness, and pigmentation (<i>Dusaran & Cabarles, 2005</i>). According to the Philippine Statistics Authority (PSA), As of January 2021, more than 42.9 % of the total inventories of chicken in the Philippines consist of the pure native and upgraded native chicken which are mostly raised under backyard condition. Demand for poultry meat was increasing every year according to PSA the projected demand will be around 1.4 million metric tons in 2022. For many years, the native chicken	

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production in Western Visayas has been a common livelihood for many farmers. Native chickens are raised essentially as scavengers in the backyard and farmyards. There is still considerable potential for continuation and expansion of this small scale approach in many communities. Farmers prefer native chickens because it does not require special care and feeds but they have nutritious products that are free from toxins compared to commercial broilers. Native chicken does not require high cost of production and capitals and they could thrive under rugged conditions (*Dusaran & Cabarles, 2005*).

Native chickens are very popular as having the tasty meat, ability to utilize farm by products and resistant to diseases. Despite of native chicken's potential to resist many diseases. However, Newcastle's Disease (ND) has been reported to have significantly affect poultry production in the Philippines. Newcastle's disease is a highly contagious disease of birds caused by a para-myxo virus. In the Philippines, the disease occurs all year round in all 17 regions, including the region of Western Visayas wherein poultry raising is an important means of livelihood for small scale farmers. Thus, ND has been a serious problem as it causes major loss to the farmers.

Newcastle disease (ND) is an acute and highly contagious virus infection that can affect most bird species. The disease is endemic in many parts of the world and causes big economic losses due to high mortality and reduced production. In rural areas the disease can kill up to 80% of unprotected poultry and is thereby one of the biggest constraints to village poultry production and a considerable restrict of rural development (*Alexander, 2004*). In the Philippines, the disease occurs all year round in all 81 provinces, including the Province of Bohol wherein poultry raising is an important mean of livelihood for smallhold farmers. Poultry population in the province comprised largely of native chickens produced by smallholders for about 90%. Thus, ND has been a serious problem as it causes major losses to these farmers (*Tapdasan, et al., 2013*).

Newcastle disease still has no treatment discovered once it hits the flocks of native chickens, thus causes a great losses to our farmers particularly backyard raisers. Only vaccines are used to prevent the entry of this disease in many poultry farms. To find alternative solution to combat this chicken disease, a study on the use of serpentina plant powder was conducted.

Serpentina is also renowned medicinal herb for its different pharmacological activities such as antibacterial, antioxidant and antifungal which also attributed to livestock and poultry nutrition (*Okwu, , 2004*). Serpentina (*Andrographis paniculata*) plant is a medicinal herb commonly growing among the tropical countries in Asia. This plant is also known as King of Bitters and known to treat inflammation, upper respiratory tract infection, diabetes, hypertension, flu, diarrhea, and liver problems (<https://www.pchrd.dost.gov.ph>, 2022). Serpentina plant extract shows the metabolites like amino acids, alkaloids, carbohydrates, flavonoids, phenols, terpinoids, oils, Saponins, sterols and lipids (*Ratnam, 2020*). It has been reported that serpentina plant had shown great potential in increasing the immune response of chicken against many different diseases. However, there is insufficient information on the effect of serpentina plant on the parasites and Newcastle's diseases in native chicken hence, this study was conducted.

Generally, the study was conducted to evaluate the effect of powdered serpentina plant (*Andrographis paniculata*) administration on the Newcastle disease titer level of Darag native chicken (*Gallus gallus domesticus*). Specifically it aims to measure Newcastle disease of titer result after 2, 4, 8 and 12-weeks post treatment of powdered serpentina plant at different levels of treatments, evaluate the effect of the powdered plant in the Serum Glutamic Oxaloacetic Transaminase (SGOT) and Serum Glutamic Pyruvic Transminase (SGPT) profile of Darag native chicken and find out the effect of the powdered plant in the productive performance of Darag native chicken.

(11) Materials and Methods or Methodology

Materials

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The materials that were used in the study includes the following:

Two hundred twenty five (225) heads of naturally infested Darag native chickens of both sexes, with ages ranging from 2 months to 4 months old; poultry house with rearing pens, serpentina plants, blender, knife, commercial feeds, weighing scale, feeders, waterers, old newspapers, electric microscope, test tubes, test tube racks, slides, Mc Master slide, sugar solutions, plastic bags, commercial drugs, syringes, micro tube, test tube with EDTA, centrifuge, artificial egg incubator, egg trays, egg color fan, camera, labeling materials and record book.

Methods

Experimental Treatments. The study were composed of two factors. Factor A are the kind of Serpentina parts and commercial drugs and Factor B are the different level/ dosages. Factor A includes: A1 – Control, A2 – Serpentina leaves powder , A3 – Serpentina fruit with seeds powder, A4 – Commercial Dewormer and A5- Commercial Coccidiostat. Factor B were: B1 – 1.0 g/kg BW, B2 – 1.5 g/kg BW and B3 – 2 g/kg BW of Darag native chickens.

Experimental Lay-out and Design. A 5x3 factorial experiment in Completely Randomized Design (CRD) was used in the study. A total of 225 naturally infested experimental birds were randomly distributed to fifteen treatments, each replicated three times. Randomization was done using the draw-lots scheme. Experimental lay-out is shown in Figure 1.0.

Preparation of Housing and Equipment. A total of 45 cages with a floor space area of 1 square meter each were prepared to accommodate five experimental birds to represent one experimental unit. The cages, feeders and waterers were cleaned with soap and water and disinfected using common commercial disinfectant available in the market. The cages and equipment were allowed to air – dry for one week to ensure that the active ingredients were evaporated. The cleanliness of the cages and other materials were also maintained.

A1B 1	A3B 1	A4B 2	A2B 1	A5B 1	A3B 3	A5B 3	A1B 2	A2B 2	A1B 3	A4B 3	A2B 3
A1B 3	A4B 3	A2B 3	A3B 2	A4B 1	A2B 2	A3B 3	A5B 2	A3B 1	A5B 3	A1B 1	A1B 2
H A L L W A Y											
A2B 1	A5B 1	A3B 1	A4B 3	A1B 1	A4B 2	A5B 2	A2B 3	A4B 1	A3B 3	A4B 2	
A1B 2	A3B 2	A5B 3	A2B 2	A4B 1	A5B 2	A1B 3	A5B 1	A3B 2	A2B 1		

Figure 1.0. Experimental lay-out and distribution of experimental birds in pen arranged in a 5 x 3 factorial experiment.

Legend:

Factor A – Kind of Serpentina plant preparation and Commercial Drugs

A1 – Control

A2 – Serpentina leaves powder

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A3 – Serpentina fruit with seeds powder
A4 – Commercial Dewormer
A5 – Commercial Coccidiostat

Factor B – Dosage
B1 – 1.0 g/kg BW*
B2 – 1.5 g/kg BW
B3 – 2.0 g/kg BW
*BW – body weight

Procurement and Quarantine of Experimental Animals. Experimental birds (pure Darag native chickens) were procured from the WVSU Darag Native Chicken Research Project. The birds were quarantined for two weeks in the quarantine house away from the research site. They were also given with multivitamins and electrolytes to recover from stress during transport. After the quarantine period, they were transferred to their assigned pens for data gathering.

Feeds and Feeding. Commercial grower ration (mash or crumble) were given for the whole duration of the study for Phase I and chicken breeder feeds for Phase II. Experimental birds were fed twice daily; one in the morning and one in the afternoon.

Provision of Water. The experimental birds were provided with fresh and clean tap water every day. Waterers were cleaned and replaced when found dirty. Antibiotic and vitamins were given during the stressful conditions particularly before and after body weights were taken and during blood collection.

Health Management of Experimental Birds. Prior to the start of the study, all experimental birds were given with Newcastle disease vaccine to avoid disease outbreak. However, they were not given with any dewormer in preparation of the study. During stress days such as weighing, blood collection, and in rainy seasons, they were given with anti-stress supplements.

Collection and Selection of Serpentina Plants. Serpentina plants were gathered both from the Darag Native Chicken Research area and some other parts of Dumarao and Dao, Capiz. The plants were collected, leaves and fruits were cut in their stems, washed and sundried to produce powdered serpentina leaves.

Administration of Treatments. Powdered serpentina plants and commercial drugs were weighed based on the amount/dosage required and were mixed to feeds (mash form). Treatments were administered and were repeated 14th day after the first administration and follow up after 2 months. Fasting of birds were done by skipping one meal to ensure that all powdered serpentina and commercial drugs were consumed. Administrations of treatments were done in the afternoon feeding. The treatments were given at dosages of 1.0, 1.5 and 2.0 grams per kilogram body weight of experimental birds.

Data Gathering Procedures

New Castle Disease Titer Test. Blood samples (serum) taken from the experimental animals were brought to DA RFO VII in Cebu City in their Vaccine Laboratory for the HI Titer test. Pre-treatment tests were done before the administration of serpentina and commercial drugs and post-treatment at 2, 4, 8, and 12 weeks after treatment administration.

Serum Biochemical Parameter Analysis. To determine the serum biochemical parameters of experimental birds, pre treatment analysis were done prior treatment administration and post treatment at 12 weeks after treatment. These blood values includes biochemical values in terms of Serum glutamic oxaloacetic transaminase (SGOT) and Serum glutamic pyruvic transaminase (SGPT). Blood samples were withdrawn from the experimental birds and this were examined Biofera Diagnostic Laboratory in Roxas City for the hematological and serum biochemical test.

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Productive Performance. Productive performance were determined to measure the effect of serpentina plant powder by determining the gain in weight, feed consumption, feed conversion ratio, mortality and morbidity rate and the net profit of birds (Ozaraga, 2012). Below are the procedures in data gathering of productive performance of experimental animals:

Feed consumption. The birds' feed intake (grams) were determined daily. Feeds were weighed using a 1000-gram capacity weighing scale. In the morning, left over and wastage were collected and subtracted to the amount given the previous day. The difference represents the amount of feed consumed.

Gain in weight. The gain in weight (grams) were determined by weighing the experimental birds at the start and at the end of the experimental period. The initial weight obtained were subtracted from the final weight of the birds. Weighing of the birds were done in the morning before feeding to minimize error on the 7th, 14th, 21st and 28th day, 2nd month and 3rd month post treatment. The gain in weight were expressed in grams (Abella, 2013).

Feed Conversion Ratio. FCR of the experimental birds were computed based on the average feed consumption over the average gain in weight of the chickens.

Net Profit. The net profit were calculated by subtracting the total expenses from the total sales obtained from the live chickens. This were done by computing all expenses incurred in the study and were subtracted from the total sales of the birds after 90 days of experimental period.

Data Analysis Procedures

The data gathered were the level of the Newcastle Titer of Darag native chicken before and after given with powdered serpentina plant and serum biochemical profile (SGOT and SGPT) count of Darag native chicken before and after 12 weeks post treatment given with powdered serpentina plant. Data were taken from the result given by the Vaccine Laboratory of DA RFO 7 and the laboratory result provided by Biofera Laboratory Diagnostic Laboratory.

Data collected were analyzed using the Analysis of Variance (ANOVA). Differences in treatment means were compared using the Duncan Multiple Range Test (DMRT). The result were interpreted at five percent level of significance.

(12) Results and Discussions

Table 1.0 presents the geometric mean titer of Darag native chicken as affected with different levels of Serpentina plant powder. Data revealed that Serpentina plant powder has no effect in the immune response of the Darag native chicken against New castle disease. It shows that there is a low GMT titer result after HI test for the blood serum of the Darag native chicken after 2, 4, 8 and 12 week post treatment. This results conforms to the study of Chitra (2024) on the effect of supplementation of *Andrographis paniculata* and *Curcuma longa* powder to broiler chicken. Her study revealed that broiler chicken supplemented with *Andrographis paniculata* and *Curcuma longa* powder has no effect ($P>0.05$) on the Hemagglutination Inhibition (HI) titer against Newcastle disease. The study of Shraddha et al., 2018 however, showed that broiler chickens given with methanolic extract of *Andrographis paniculata* showed a highest antibody titer result. The study of Liu et al., 2022 further shows that Muscovy ducks given with fermented and un fermented serpentina plant showed a better effects that improved growth, immune status, intestinal morphology and intestinal microbiota composition of the ducks making it a potential alternative to antibiotic in poultry production. This result was the effect of phytochemicals compound of the *Andrographis paniculata*, the phenols and flavonoids which are known to have beneficial importance in immunomodulation.

Table 1.0 Geometric mean titer of HI Test result for Newcastle Disease of Darag native chicken as affected by different serpentina plant powder at various levels.

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Kind of Serpentina Plant Preparation	Level of Serpentina Plant Powder (g/kg BW)	Period				
		Pre	Week 2 ^{ns}	Week 4 ^{ns}	Week 8 ^{ns}	Week 12 ^{ns}
Control		3.44	0.67	0.45	0.00	0.89
Serpentina Leaves Powder	1.0	4.00	0.00	0.00	0.00	1.33
	1.5	3.33	1.33	0.67	0.00	1.33
	2.0	3.67	0.00	0.00	0.00	1.33
Serpentina Fruit and Seeds Powder	1.0	3.67	0.67	0.00	0.00	2.00
	1.5	4.33	0.67	0.00	0.67	0.67
	2.0	4.67	1.33	0.00	0.00	0.67
Commercial Drug		3.61	0.78	0.50	0.11	1.06

Effect of Serpentina Plants (*Andrographis paniculata*) on the Serum Glutamic Pyruvate Transaminase (SGPT) of Darag Native Chickens

Table 2.0 shows the serum glutamic pyruvic transaminase (SGPT) of Darag native chicken as affected by different serpentina plant parts powder at various levels. The study reveal that there is no significant difference for the effect of the different serpentina plant powder in the SGPT of the experimental birds. Data further shows that the SGPT count of Darag native chickens ranged from 15.00 – 19.00 after 90 days' post treatment which is a little bit lower in the pretreatment period where in the total SGPT count range from 13.33 – 20.60. This results are lower than the normal SGPT count which is 23. 85 28 iu/l (Rahmania, 2022). This means that Serpentina plant powder did not increase the SGPT count of the experimental birds. This shows that experimental birds have a healthy liver.

Table 2.0 The serum glutamic pyruvic transaminase (SGPT) of Darag native chicken as affected by different serpentina plant parts powder at various levels.

Kind of Serpentina Plant Preparation	Level of Serpentina Plant Powder (g/kg BW)	Period	
		Day 0 ^{ns}	Day 90 ^{ns}
Control		19.89	16.44
Serpentina Leaves Powder	1.0	13.33	19.00
	1.5	19.33	15.00
	2.0	20.33	17.00
Serpentina Fruit and Seeds Powder	1.0	18.33	17.67
	1.5	16.67	15.33
	2.0	20.67	17.00
Commercial Drugs		19.76	20.61

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Effect of Serpentina Plants (*Andrographis paniculata*) on the Serum Glutamic Oxaloacetic Transaminase (SGOT) of Darag Native Chickens

Table 3.0 shows the serum glutamic oxaloacetic transaminase (SGOT) of Darag native chicken as affected by different serpentina plant parts powder at various levels. The study reveal that there is no significant difference for the effect of the different serpentina plant powder in the SGOT of the experimental birds. Data further shows that the SGOT count of Darag native chickens ranged from 184.67 – 213.00 after 90 days' post treatment which is a little bit lower in the pretreatment period where in the total SGOT count range from 226 – 278.33. This results are conforms to the normal SGOT count of a chicken which is 37. 8 to 311 iu/l (Rahmania, 2022). This means that Serpentina plant powder did not increase the SGOT count of the experimental birds. This shows that liver of the experimental birds was not damaged by giving them the Serpentina plant powder. The andrographolide shows hepatoprotective (Mishra et al., 2017) against various types of liver damage in which the liver is directly affected when a person diagnosed with diabetes (Widjajakusuma 2019). Result of the study showed that Serpentina plant powder did not affect the liver of the experimental birds. Because the presence of damaged and dead liver cells results in increased migration of the amine group transfer enzymes (transaminase) of both Serum Glutamate Oxaloacetate Transaminase (SGOT) and Serum Glutamate Pyruvate Transaminase (SGPT) into the blood circulation system (Rahmania, 2022).

Table 3.0 The serum glutamic oxaloacetic transaminase (SGOT) of Darag native chicken as affected by different serpentina plant parts powder at various levels.

Kind of Serpentina Plant Preparation	Level of Serpentina Plant Powder (g/kg BW)	Period	
		Day 0 ^{ns}	Day 90 ^{ns}
Control		237.55	222.56
	1.0	262.33	197.00
	1.5	226.00	193.00
Serpentina Leaves Powder	2.0	278.33	213.00
	1.0	249.33	184.67
	1.5	259.33	196.00
	2.0	276.33	215.67
		261.89	207.11
Commercial Drugs			

Effect of Serpentina Plants (*Andrographis paniculata*) on the Productive Performance of Darag Native Chickens

Table 4.0 shows the serum glutamic oxaloacetic transaminase (SGOT) of Darag native chicken as affected by different parts of serpentina plant powder at various levels. Data shows that Darag native chicken given with Serpentina leaves powder at 1 g/kg bodyweight obtain the highest gain in weight and best conversion ratio. The result further showed that birds given with serpentina leaves at 1, and 1.5 g/kg bodyweight and serpentina fruits and seeds powder at 2g/kg bodyweight has a similar effect to the gain in weight of those birds given with commercial drugs. Data further reveal that Darag native chicken given with Serpentina leaves powder obtain the higher net profit compare to those given with

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fruits and seeds powder. This results conform to the study of Liu et al., 2022 that ducks given with fermented serpentina plants give the lowest feed conversion ratio and better growth of ducks. Study of Chitra (2024) further showed that broiler chickens supplemented with *Andrographis paniculata* and *Curcuma longa* powder improved the growth rate and feed conversion ratio of broilers chicken. Chitra further

Table 4.0 Productive performance of Darag native chicken as affected by different serpentina plant powder at various levels.

Kind of Serpentina Plant Preparation	Level of Serpentina Plant Powder (g/kg BW)	Performance			
		Feed Consumption (g)	Gain in Weight (g)	Feed Conversion Ratio	Net Profit (Php)
Control		5329.49	1190.00 ^{bc}	4.54 ^{bc}	97.41
Serpentina Leaves Powder	1.0	5337.62	1447.00 ^a	3.69 ^a	107.06
	1.5	5338.71	1255.00 ^{ab}	4.29 ^{bc}	115.62
	2.0	5341.20	1127.00 ^c	4.75 ^c	82.56
Serpentina Fruit and Seeds Powder	1.0	5336.93	1092.00 ^c	4.89 ^c	78.58
	1.5	5338.62	1132.33 ^b	4.74 ^c	55.21
	2.0	5340.01	1285.33 ^{ab}	4.16 ^b	51.42
Commercial Drug		5342.81	1307.39 ^{ab}	4.10 ^b	142.92

(13) Conclusions and Recommendations

Based on the results of the study, the researcher concludes that Serpentina (*Andrographis paniculata*) plant powder has no effect in the immune response of Darag native chickens against Newcastle disease. The powdered plants can lower the SGPT and SGOT of Darag native chickens which shows its hepatoprotective property. The serpentina plant powder can give a better growth rate, feeds conversion ratio and net profit to Darag native chickens.

Based on the findings of the study, the researcher recommends the use of serpentina plant powder as supplement to Darag native chicken to improved growth rate and feed conversion ratio. Explore another study of using different formulation of the serpentina plants (extract or liquid form) for its immunomodulatory effect and as natural antibiotic property to other animals and other chicken diseases.

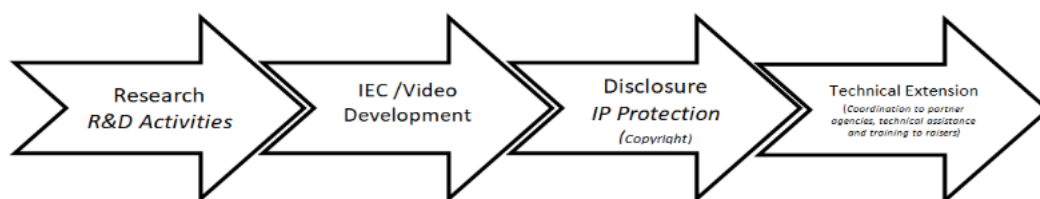
(14) Output

Expected Output	Details/ Activities Conducted	Target
People Services	Train/ orient native chicken growers on the use of serpentina plant powder as supplements to	30 raisers

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	their native chickens to help in the improvement of the growth rate and production.	
Products	Production of powdered serpentina plant powder	10 kilos
Places and Partnerships	Forge linkages and partnerships to LGU's and other SUC's for future collaboration of the further study of serpentina plant	2
Publications	Develop/ produce Information, Education and Communication (IEC) and promotional materials for preparation and utilization of Serpentina plant.	2
Patents/IP	File 2 copyrights for the developed IEC materials.	2
Policy	Develop one policy inputs to for the utilization of serpentina plant powder as feed supplements.	1
Impact	Help native chicken raisers to improved native chicken production	

(15) Technology Transfer Activity Conducted



(16) Economic Analysis (This should present the direct cost of project implementation, expenses in the production of technology, revenues generated from technology produced, return on investment (ROI) and payback period.

(17) Acknowledgments

The researcher would like to thank the CapSU RDE and LRDC for funding the study.

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